corrosion inhibitors, which help to keep the cooling system clean and undamaged. Since these inhibitors get used up over time, you should change the antifreeze annually in order to renew them, or at the first signs of rust in the cooling system.

Bleeding a Cooling System. When you start an engine after any part of the cooling system has been drained, you must check the flow of coolant, particularly on the raw-water side. If a water pump is airbound, or water fails to circulate for any other reason, the vanes will strip off rubber impellers in seconds; cylinder heads can crack in minutes.

Bleed an air-bound cooling system just as you would a fuel system, starting from the water source and working toward the pump. Once the pump is primed, it should push the coolant through the rest of the system. However, on occasion air-locks can form in hoses, stalling out the water flow—the long hose runs associated with the heat exchangers in domestic water heaters are particularly prone to this. If this is a problem, progressively loosen hose connections around the system, allowing water to vent out of each connection before retightening.

Hoses and Hose Connections. Suction-side hoses to water pumps need to be of the non-collapsing type, preferably with an internal *stainless steel* spring. In time, hoses that are subjected to heat soften and bulge, or crack, and need replacing.

Hose clamps must be of stainless steel, including the worm screw. All hose connections below the waterline need to be double-clamped. Once a year, you should loosen hose clamps and inspect the band where it has been *inside the screw housing*—the bands are prone to failure from crevice corrosion at this point.

Thermostats. Thermostats are generally very reliable, but occasionally do fail, normally in the closed position. Failure leads to overheating. The thermostat housing is almost always near the top of the engine, at the front (the crankshaft pulley end), with one or more cooling hoses coming out of it (see Figure 6-3 and Figure 6-4). Generally, it is held down with a couple of bolts. Once these are removed, gently pry up the housing. The thermostat can be lifted out and tested as described on page 72. Thermostats are rela-



Figure 6-3. *Thermostat housing on a Volvo MD17C.*



Figure 6-4. *Thermostat removed from a Volvo MD17C.*